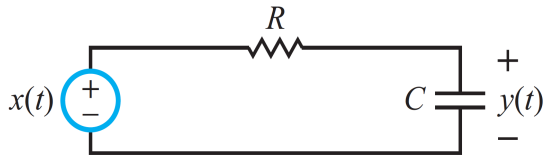
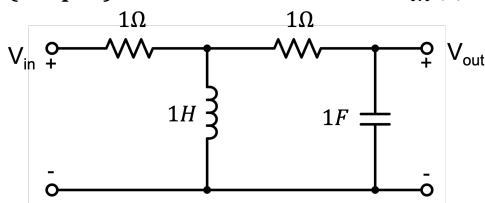


1. (30 pts) For the following RC circuit, calculate the following outputs for $x(t) = 25 \cos(3t)u(t)$ when $R = 250k\Omega$, $C = 1\mu F$, and an initial capacitor voltage of 1V.



- a. Zero-input response
 - b. Zero-state response
 - c. Transient response
 - d. Steady-state response
 - e. Natural response
 - f. Forced response
2. (30 pts) Given an LTI system with an impulse response $h(t) = 2\delta(t) + e^{-4t} \sin(2t) u(t)$, determine the following:
- a. $\mathbf{H(s)}$
 - b. $\hat{\mathbf{H}}(\omega)$
 - c. Poles and zeros of the system
 - d. LCCDE description of the system
 - e. Output response $y(t)$ to input $x(t) = 2te^{-5t}u(t)$. Assume zero initial conditions.
3. (40 pts) For the circuit below, $V_{in}(t) = 1$ V for $t < 0$, and $V_{in}(t) = (1 - e^{-t})u(t)$ V for $t > 0$

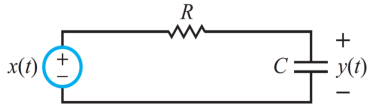


- a. (5 points) Calculate the initial capacitor voltages and inductor currents
- b. (5 points) Draw the $\mathbf{s}$ -domain representation of the circuit.
- c. (5 points) Find the $\mathbf{s}$ -domain representation of the input $V_{in}(t)$ for $t > 0$.
- d. (5 points) Write mesh current equations in the $\mathbf{s}$ -domain that could be used to find the current through the capacitor. Use these equations to fill in the entries in the Cramer's method matrices shown below. Here, $\mathbf{I_1}$ and $\mathbf{I_2}$ represent the mesh currents, and $\mathbf{V_1}$, $\mathbf{V_2}$ respectively represent voltage sources.

$$\begin{pmatrix} \mathbf{A_1} & \mathbf{A_2} \\ \mathbf{A_3} & \mathbf{A_4} \end{pmatrix} \begin{pmatrix} \mathbf{I_1} \\ \mathbf{I_2} \end{pmatrix} = \begin{pmatrix} \mathbf{V_1} \\ \mathbf{V_2} \end{pmatrix}$$

- e. 10 points) Find the output $\mathbf{V_{out}(s)}$ in the Laplace domain.
- f. (10 points) Find the output $v_{out}(t)$ in the time domain.

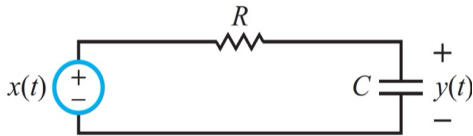
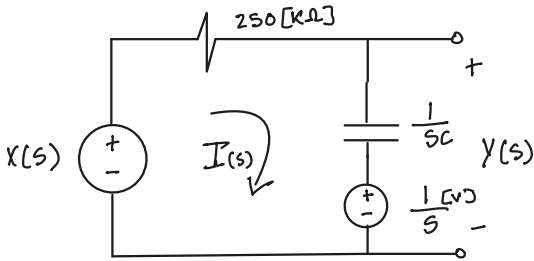
1. (30 pts) For the following RC circuit, calculate the following outputs for $x(t) = 25 \cos(3t)u(t)$ when $R = 250k\Omega$, $C = 1\mu F$, and an initial capacitor voltage of 1V.



- Zero-input response
- Zero-state response
- Transient response
- Steady-state response
- Natural response
- Forced response

Laplace Domain:

$$X(s) = \frac{25s}{(s^2 + 9)}$$



- Zero-input response
- Zero-state response
- Transient response
- Steady-state response
- Natural response
- Forced response

$$Y(s) = \frac{-16}{s+4} + \frac{10e^{-j36.87}}{s-3j} + \frac{10e^{j36.87}}{s+3j} + \frac{1}{s+4}$$

$$Y(t) = -15e^{-4t} + 20 \cos(3t - 36.87)u(t)$$

↑ C.)
transient response
or
y natural
E)

↑ D.)
steady state
or
y forced
F)

loop:

$$-X(s) + RI(s) + \frac{1}{sC}I(s) + \frac{V_c(0^-)}{s} = 0$$

$$[R + \frac{1}{sC}]I(s) = X(s) - \frac{V_c(0^-)}{s}$$

$$I(s) = \frac{X(s) - \frac{V_c(0^-)}{s}}{[R + \frac{1}{sC}]}$$

$$I(s) = \frac{X(s)}{[R + \frac{1}{sC}]} - \frac{V_c(0^-)}{Rs + \frac{1}{C}}$$

$$I(s) = \frac{X(s)sC}{[1 + RSC]} - \frac{V_c(0^-)C}{[1 + RSC]}$$

$$Y(s) = \frac{1}{sC}I(s) + \frac{V_c(0^-)}{s}$$

$$Y(s) = \frac{1}{sC} \left[\frac{X(s)sC}{[1 + RSC]} - \frac{V_c(0^-)C}{[1 + RSC]} \right] + \frac{V_c(0^-)}{s}$$

$$Y(s) = \frac{X(s)}{1 + RSC} - \frac{V_c(0^-)}{s[1 + RSC]} + \frac{V_c(0^-)}{s}$$

$$Y(s) = \frac{X(s)}{1 + RSC} + V_c(0^-) \left[\frac{[1 + RSC] - 1}{s[1 + RSC]} \right]$$

$$Y(s) = \frac{X(s)}{1 + RSC} + V_c(0^-) \left[\frac{RSC}{s[1 + RSC]} \right]$$

$$Y(s) = \frac{X(s)}{1 + RSC} + V_c(0^-) \left[\frac{RC}{[1 + RSC]} \right]$$

- Zero-input response
- Zero-state response

$$Y(s) = \left[\frac{1/4}{[1 + 1/4s]} \right]$$

$$Y(s) = \frac{1}{(s+4)}$$

$$a.) y(t) = 1e^{-4t}u(t)$$

$$\frac{25s}{s^2 + 9} \div \frac{1 + 1/4s}{1}$$

$$Y(s) = \frac{100s}{(s^2 + 9)(s+4)} = \frac{100s}{(s+4)(s-3j)(s+3j)} = \frac{A}{(s+4)} + \frac{B}{(s-3j)} + \frac{C}{(s+3j)}$$

$$\frac{100s}{(s^2 + 9)} \Big|_{s=-4} = \frac{100(-4)}{16 + 9} = \frac{-400}{25} = -16$$

$$\frac{100s}{(s+3j)(s+4)} \Big|_{s=3j} = \frac{300j}{(6j)(3j+4)} = \frac{50}{(4+3j)}$$

$$\frac{50}{4+3j} \cdot \frac{4-3j}{4-3j} = \frac{200-150j}{16+9} = 8-6j$$

$$A = -16$$

$$B = 8-6j$$

$$C = 8+6j$$

$$\frac{-16}{s+4} + \frac{8-6j}{s-3j} + \frac{8+6j}{s+3j}$$

$$\frac{-16}{s+4} + \frac{10e^{-j36.87}}{s-3j} + \frac{10e^{j36.87}}{s+3j}$$

$$b) -16e^{-4t} + 20 \cos(3t - 36.87)u(t)$$

2. (30 pts) Given an LTI system with an impulse response $h(t) = 2\delta(t) + e^{-4t} \sin(2t) u(t)$, determine the following:

- $H(s)$
- $\hat{H}(\omega)$
- Poles and zeros of the system
- LCCDE description of the system
- Output response $y(t)$ to input $x(t) = 2te^{-5t}u(t)$. Assume zero initial conditions.

$$H(s) = 2 + \frac{2}{(s+4)^2 + 4} \quad X(s) = \frac{2}{(s+5)^2}$$

$$H(s) = 2 + \frac{2}{s^2 + 8s + 16 + 4}$$

$$H(s) = \frac{2(s^2 + 8s + 20) + 2}{s^2 + 8s + 20}$$

$$a.) H(s) = \frac{2s^2 + 16s + 42}{s^2 + 8s + 20} \quad \frac{Y(s)}{X(s)}$$

$$b.) H(\omega) = \frac{2(j\omega)^2 + 16(j\omega) + 42}{(j\omega)^2 + 8j\omega + 20}$$

$$H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau.$$

$$X(s) = \mathcal{L}[x(t)] = \int_0^{\infty} x(t) e^{-st} dt,$$

$$\frac{-16 \pm \sqrt{16^2 - 4(2)(42)}}{4}$$

$$\frac{-8 \pm \sqrt{8^2 - 4(1)(20)}}{2}$$

$$c.) \text{ Zeros} = -4 \pm \sqrt{5} j$$

$$\text{Poles} = -4 \pm 2j$$

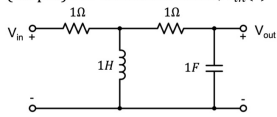
$$H(s) = \frac{Y(s)}{X(s)} \quad H(s)X(s) = Y(s)$$

$$D.) H(s) = 2s^2 + 16s + 42 \quad X(s) = s^2 + 8s + 20 \quad Y(s)$$

$$E.) Y(s) = \frac{2}{(s+5)^2} \cdot \left(\frac{2s^2 + 16s + 42}{s^2 + 8s + 20} \right)$$

$$Y(s) = \frac{4(s^2 + 8s + 21)}{(s+5)^2 (s^2 + 8s + 20)}$$

3. (40 pts) For the circuit below, $V_{in}(t) = 1$ V for $t < 0$, and $V_{in}(t) = (1 - e^{-t})u(t)$ V for $t > 0$

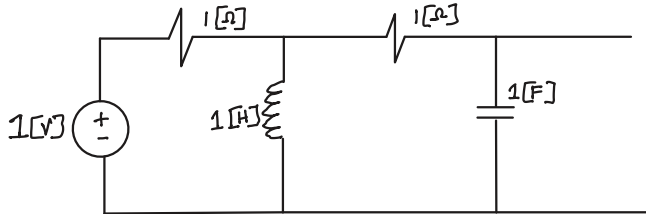
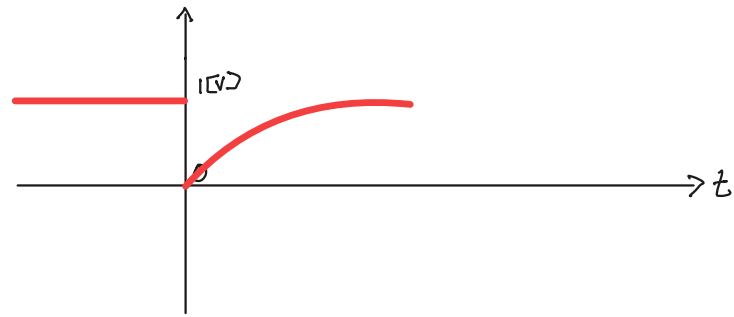


- (5 points) Calculate the initial capacitor voltages and inductor currents
- (5 points) Draw the s-domain representation of the circuit.
- (5 points) Find the s-domain representation of the input $V_{in}(t)$ for $t > 0$.
- (5 points) Write mesh current equations in the s-domain that could be used to find the current through the capacitor. Use these equations to fill in the entries in the Cramer's method matrices shown below. Here, I_1 and I_2 represent the mesh currents, and V_1 , V_2 respectively represent voltage sources.

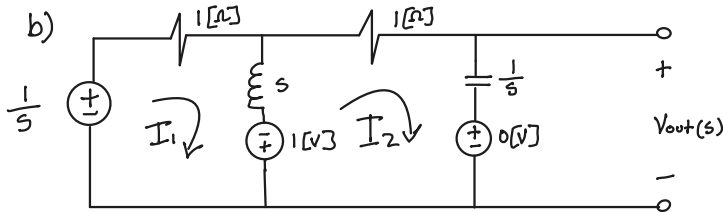
$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

e. (10 points) Find the output $V_{out}(s)$ in the Laplace domain.

f. (10 points) Find the output $v_{out}(t)$ in the time domain.



a) $i_L(0^-) = 1$ [A]
 $V_C(0^-) = 0$ [V]



$$V_{in} = \frac{1}{s} - \frac{1}{s+1}$$

c.) $(1 - e^{-t})u(t)$ [V]
 $\frac{1}{s} - \frac{1}{s+1}$

loop 1
 $-V_{in} + 1[\Omega]I_1 + s(I_1 - I_2) - 1 = 0$
 $(1+s)I_1 - sI_2 = V_{in} + 1$
 $A_1 \quad A_2$

loop 2
 $1 + s(I_2 - I_1) + 1(I_2) + 1/s I_2 = 0$
 $-sI_1 + (s+1+1/s)I_2 = -1$
 $A_3 \quad A_4$

$$\begin{bmatrix} s+1 & -s \\ -s & s+1+1/s \end{bmatrix}$$

$$\Delta = (s+1)(s+1+1/s) - s^2$$

$$\Delta = (s+1)(s+1+1/s) - s^2$$

$$\Delta = 2s+2+1/s$$

$$\Delta_2 = \begin{bmatrix} s+1 & V_{in}+1 \\ -s & -1 \end{bmatrix}$$

$$-(s+1) + s(V_{in}+1)$$

$$-(s+1) + s\left(\frac{1}{s} - \frac{1}{s+1} + 1\right)$$

$$-s-1 + 1 - \frac{s}{s+1} + s$$

$$\Delta_2 = \frac{-s}{s+1}$$

$$I_2 = \frac{-s}{s+1} \div (2s+2+1/s)$$

$$I_2 = \frac{-s}{(s+1)(2s^2+2s+1)} = \frac{A}{s+1} + \frac{Bs+C}{2s^2+2s+1}$$

$$Y(s) = \frac{-s}{(s+1)(2s^2+2s+1)}$$

$$\frac{-s}{2s^2+2s+1} \Big|_{s=-1} \frac{1}{2-2+1} \quad A=1$$

$$-s = 1(2s^2+2s+1) + Bs+C(s+1)$$

$$s=0$$

$$0 = 1 + C, \quad C = -1$$

$$s=1$$

$$-1 = 1(2+2+1) + B-1(2)$$

$$\frac{-6}{2} = \frac{(B-1)2}{2} = -3 = B-1 \quad B = -2$$

$$Y(s) = \frac{1}{s+1} + \frac{-2s-1}{2s^2+2s+1} = \frac{1}{s+1} + \frac{-2(s+0.5)}{2(s^2+s+0.5)}$$

$$Y(s) = \frac{1}{s+1} + \frac{(s+0.5)}{(s+0.5)^2 + 0.5^2}$$

$$y(t) = [e^{-t} + e^{-0.5t} \cos(0.5t)] u(t)$$